



Research paper

Fixed-time trajectory tracking control for underwater glider: theory and experiment

Xu Zhang^{a,b,c,d}, Yu Cao^{a,d}, Hexiong Zhou^{a,d,*}, Baoheng Yao^{id a,d,e}, Lian Lian^{a,d}, Zhihua Mao^e^a School of Oceanography, Shanghai Jiao Tong University, Shanghai, 200030, China^b State Key Laboratory of High-end Heavy-load Robots, Midea Group, Foshan, 528300, China^c Midea Corporate Research Center, Midea Group, Foshan, 528300, China^d State Key Laboratory of Ocean Engineering, Shanghai Jiao Tong University, Shanghai, 200240, China^e Second Institute of Oceanography, Hangzhou, 310012, China

ARTICLE INFO

Keywords:

Underwater glider
Fixed-time control
Motion control
Trajectory tracking

ABSTRACT

This paper presents a fixed-time trajectory tracking control strategy for an underwater glider (UG) operating in the vertical plane. A corresponding control architecture has been designed for embedded microcontrollers, especially suited for underwater vehicles requiring timely pitch attitude adjustment without external thrusters. This approach enables efficient and reliable implementation on the independently developed UG prototype. To assess the effectiveness of the approach, both numerical simulations and laboratory pool experiments are conducted, with performance comparisons against conventional control schemes. The results demonstrate that the proposed method exhibits enhanced robustness and rapid convergence, rendering it desirable for motion state trajectory tracking missions in UG applications involving boundary constraints.

1. Introduction

With ongoing advances in underwater glider (UG) development, their advantages in long endurance, wide-area coverage, and high-resolution sampling have become increasingly evident. UGs are now widely used for marine environmental monitoring due to their autonomous spatiotemporal sampling capabilities enabled by controllable sawtooth trajectories. These features make UGs well suited for long-term observation of key ocean processes such as thermoclines, currents, and the temperature-salinity structure of water masses. Equipped with diverse sensors, UGs serve as effective platforms for investigating typical oceanic phenomena.

In addition to observing typical ocean dynamic processes, UGs have demonstrated significant potential in a variety of constrained observation tasks. These include fine-scale thermocline profiling, collision avoidance in shallow waters with complex topography, cooperative observation by multi-body gliders (Paley et al., 2008; Leonard et al., 2010), virtual mooring and seabed monitoring (Zhou et al., 2020), autonomous docking and retrieval (Yang et al., 2016), adaptive sampling of ocean features (Feng et al., 2021), and attitude transitions in multi-modal motion (Cao et al., 2019). As task complexity and operational diversity continue to increase, greater demands are placed on the maneuverability and control precision of UGs. Consequently, the development of agile

and efficient motion control strategies has become a critical focus in this field.

The development of dynamic modeling and control methods of UG has laid a foundation for the reliable and efficient operation of ocean observation missions. Table 1 summarizes the technical features and application scenarios of representative UG control algorithms.

Although classical control methods, such as Proportional-Integral-Derivative (PID), remain widely used in practice, composite control strategies integrating advanced algorithms are emerging as a prominent trend. These approaches offer important theoretical and practical advantages for enhancing the control robustness of UG operating in complex marine environments. In area-constrained observation tasks, UGs require rapid and precise posture adjustments. However, traditional methods-such as PID, linear quadratic regulator (LQR), and various Lyapunov-based controllers-can only guarantee asymptotic stability, where system errors converge to equilibrium over an extended period. This characteristic poses limitations in time-sensitive marine observation missions, where both promptness and precision are essential.

Conventional sliding mode control (SMC) is usually based on linear sliding surface design, which is robust to disturbances and uncertainties (Shtessel et al., 2014). One of the finite-time control design methods is through terminal sliding mode control (TSMC). Compared with the asymptotic convergence characteristics of traditional sliding

* Corresponding author at: School of Oceanography, Shanghai Jiao Tong University, Shanghai 200030, China

E-mail address: zhou_hexiong@sjtu.edu.cn (H. Zhou).

Table 1
Comparative analysis of representative motion control methods for UG.

Method	Control Target	Advantages and Limitations
PID	Glide angle and turn rate (Mahmoudian and Woolsey, 2008); pitch angle and depth (Noh et al., 2011)	Simple structure and easy to implement via hardware. Requires manual parameter tuning, limited robustness, and prone to overshoot and steady-state error.
LQR	Position and velocity (Leonard and Graver, 2001); pitch angle and depth (Noh et al., 2011; Joo and Qu, 2015)	Provides optimal control based on linearized models. Performance deteriorates with significant system nonlinearity.
Backstepping	Pitch angle and depth (Cao et al., 2016)	Well-suited for strict-feedback systems. Design complexity increases with system order.
SMC	Pitch angle (Yang and Ma, 2010); posture and depth (Zhou et al., 2020)	Robust to uncertainties and disturbances. Suffers from chattering, which may excite unmodeled dynamics.
ADRC(Active Disturbance Rejection Controller)	Attitude angle (Su et al., 2019)	Disturbances can be estimated without an accurate system model. However, parameter tuning is nontrivial, and the observer performance is sensitive to measurement noise.
MPC(Model Predictive Control)	Path tracking (Tatone et al., 2009); heading and depth control (Wang et al., 2021)	Explicitly handles multiple constraints and performs rolling optimization. High computational demand and reliance on accurate prediction models may lead to suboptimal performance in real time.

mode surface, TSMC can make the system converge in a finite time by setting a nonlinear exponential sliding surface (Zak, 1988). Finite-time control theory has been identified as a promising approach to address such timeliness constraints (Yu et al., 2020). However, finite-time control algorithms have the characteristic of relying on the initial state of the system. As an extension of this theory, fixed-time control was first systematically proposed by Polyakov (2011). While retaining the rapid convergence of finite-time control, fixed-time theory guarantees convergence within a uniform upper bound that is independent of the system's initial state. This property is particularly advantageous in scenarios with unknown initial conditions or stringent time constraints.

In recent years, fixed-time control theory has gained increasing attention in underwater robotics. For example, Zhu et al. (2023) developed a fixed-time trajectory tracking controller for autonomous underwater vehicles (AUVs) using a radial basis function neural network-based disturbance observer, significantly improving both robustness and response speed. Other studies (Sun et al., 2022; Chen et al., 2023) have proposed fixed-time control schemes for AUV trajectory tracking, further enriching the control methodologies in this domain. Su et al. (2021) designed a fixed-time formation control framework for multi-AUVs, and Zuo et al. (2018) provided a comprehensive review of theoretical developments and applications of fixed-time cooperative control in multi-agent systems. Despite these advancements, most research on finite-time or fixed-time control for underwater vehicles remains at the theoretical or simulation stage, with limited experimental validation or engineering implementation. Specifically, in the UG domain, there is a lack of systematic investigation and practical integration of finite/fixed-time convergence strategies. As ocean observation tasks become increasingly complex, the development of advanced control algorithms with both rapid convergence and strong robustness is becoming a critical direction to enhance UG responsiveness and autonomous decision-making under constrained mission conditions. Similarly, many UG control works (Zou et al., 2024; Wan et al., 2022; Wang et al., 2025; Zhang et al., 2023) only focus on theoretical analysis and simulation, lacking experimental verification in UG prototypes. The main reason is that many algorithms are difficult to deploy on the control board, especially on the STM32 microcontroller used in this paper.

Inspired by the above studies, this study focuses on using UG to complete zigzag trajectory tracking control in the vertical plane under constrained tasks. Due to the constrained observation tasks, such as shallow water collision avoidance or thermocline boundary tracking, high requirements are placed on the responsiveness and robustness of the control system. In order to solve the observation tasks based on UG under constrained boundaries, a fixed-time trajectory tracking control scheme

is proposed. The main contributions of this paper are summarized as follows:

- A fixed-time trajectory tracking control method is developed for UG operating in the vertical plane. Compared with asymptotic (Yang and Ma, 2010; Zhou et al., 2020) or finite-time control (Feng et al., 2002, 2013) approaches, the proposed method not only ensures that the system state converges within a finite time, but also its convergence time does not depend on the initial state, and has lower requirements on the initial state of the system.
- The proposed control algorithm features a lightweight structure with low computational complexity, making it readily implementable on the onboard main controller of the UG prototype, with control parameters that are straightforward to tune and adapt.
- Comprehensive laboratory pool experiments were conducted on the self-developed glider prototype under multiple control schemes. The results demonstrate that the proposed fixed-time control approach effectively ensures tracking accuracy and fixed-time convergence.

The remainder of this paper is organized as follows: Section 2 introduces the dynamic model of UG in the vertical plane. Section 3 presents the design of the fixed-time trajectory tracking control strategy along with its stability analysis. Section 4 provides simulation and experimental results to validate the effectiveness and fixed-time convergence of the proposed approach. Section 5 concludes the paper and outlines future research directions.

2. Preliminaries and problem formulation

2.1. Notations

The following notations are used throughout the paper.

For a vector $\eta \in \mathbb{R}^n$, the norm η stands for the Euclidean norm, i.e., $\|\eta\| = \sqrt{\eta^\top \eta}$. For a matrix $\Xi \in \mathbb{R}^{n \times n}$, $\lambda_{\min}(\Xi)$ and $\lambda_{\max}(\Xi)$ represent the minimum and maximum eigenvalues of matrix Ξ respectively. The symbol $\text{diag}(\odot)$ stands for the diagonal matrix.

To facilitate controller design and prove stability, the following lemmas are given.

Lemma 1. For any $x \in \mathbb{R}^n$ and any $\zeta > 0$, the following inequality holds:

$$0 \leq \|x\| - x^\top \tanh\left(\frac{x}{\zeta}\right) = 0.2785n\zeta, \quad (1)$$

where $\tanh\left(\frac{x}{\zeta}\right) = [\tanh\left(\frac{x_1}{\zeta}\right), \tanh\left(\frac{x_2}{\zeta}\right), \dots, \tanh\left(\frac{x_n}{\zeta}\right)]^\top$.

Proof. Following the lemma in Polycarpou (1996), one can derive:

$$0 \leq |x_i| - x_i \tanh\left(\frac{x_i}{\zeta}\right) = 0.2785\zeta (\forall i = 1, 2, \dots, n). \quad (2)$$

Using the norm property of vectors:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{[x_1, x_2, \dots, x_n]^T [x_1, x_2, \dots, x_n]}, \quad (3)$$

combining Eqs. (2) and (3), it follows that:

$$\begin{aligned} \|\mathbf{x}\| &\leq \sqrt{(|x_1| + |x_2| + \dots + |x_n|)^2} \leq |x_1| + |x_2| + \dots + |x_n| \\ &\leq \sum_{i=1}^n \left[x_i \tanh\left(\frac{x_i}{\zeta}\right) + 0.2785\zeta \right] \\ &= \mathbf{x}^T \tanh\left(\frac{\mathbf{x}}{\zeta}\right) + 0.2785n\zeta. \end{aligned} \quad (4)$$

This completes the proof of Lemma 1. $\square$

Lemma 2. (Liu et al., 2025) Consider the nonlinear system:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t)), \quad \mathbf{f}(\mathbf{0}) = \mathbf{0}, \quad \mathbf{x}(0) = \mathbf{x}_0 \quad (5)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the system state, $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a nonlinear function, and $\mathbf{x}_0 \in \mathbb{R}^n$ is the initial condition. Assume there exists a positive definite Lyapunov function $V(\mathbf{x}) = \mathbf{x}^T \mathbf{x}$ such that:

$$\begin{cases} V(\mathbf{x}) > 0, \\ \dot{V}(\mathbf{x}) \leq -2c_1 V^{\frac{\aleph_1+1}{2}}. \end{cases} \quad (6)$$

in which $\aleph_1 = \frac{m+n}{2} + \left(\frac{m-n}{2}\right) \text{sign}(\|\mathbf{x}\| - 1)$, with constants $m > 1, 0 < n < 1$, and $c_1 > 0$. Then, the system achieves fixed-time convergence, and the settling time satisfies:

$$T(\mathbf{x}_0) < T_{\max 1} = \frac{1}{c_1(m-1)} + \frac{1}{c_1(1-n)}. \quad (7)$$

Lemma 3. (Liu et al., 2025) Consider the system (5) with a Lyapunov function $V(\mathbf{x}) = \mathbf{x}^T \mathbf{x}$ that satisfies:

$$\begin{cases} V(\mathbf{x}) > 0, \\ \dot{V}(\mathbf{x}) \leq -2c_2 V^{\frac{\aleph_2+1}{2}} + l. \end{cases} \quad (8)$$

where $\aleph_2 = \frac{m+n}{2} + \left(\frac{m-n}{2}\right) \text{sign}(\|\mathbf{x}\| - 1)$, $m > 1$, $0 < n < 1$, and $c_2 > 0$. If $l > 0$ and $\bar{\kappa} \in (0, 1)$, then the system achieves fixed-time stability, and the convergence time satisfies:

$$T(\mathbf{x}_0) < T_{\max 2} = \frac{1}{c_2 \bar{\kappa}(m-1)} + \frac{1}{c_2 \bar{\kappa}(1-n)} \quad (9)$$

Moreover, the system state converges to the compact set:

$$\mathcal{F} = \left\{ \mathbf{x} \left| V(\mathbf{x}) \leq \min \left\{ \left(\frac{l}{2(1-\bar{\kappa})\aleph_2} \right)^{\frac{2}{m+1}}, \left(\frac{l}{2(1-\bar{\kappa})\aleph_2} \right)^{\frac{2}{n+1}} \right\} \right. \right\}. \quad (10)$$

2.2. Underwater glider model

This paper focuses on the zigzag motion of UG in the vertical plane. The simplified dynamic model used in Zhang et al. (2024) is given as follows:

$$\begin{cases} \dot{x} = u \cos \theta + w \sin \theta, \\ \dot{z} = -u \sin \theta + w \cos \theta, \\ \dot{\theta} = q, \\ \dot{q} = \frac{1}{I_{f2}} \left[(M_{f3} - M_{f1} + K_M)uw - m_b r_{b1} g \cos \theta - m_p g (r_{p1} \cos \theta + R_p \sin \theta) \right. \\ \quad \left. + K_{M0}u^2 + K_q qu^2 \right], \\ \dot{u} = \frac{1}{M_{f1}} [-K_{D0}u^2 + K_{L0}uw - m_b g \sin \theta], \\ \dot{w} = \frac{1}{M_{f3}} [-K_{L0}u^2 - (K_L + K_{D0})uw + m_b g \cos \theta]. \end{cases} \quad (11)$$

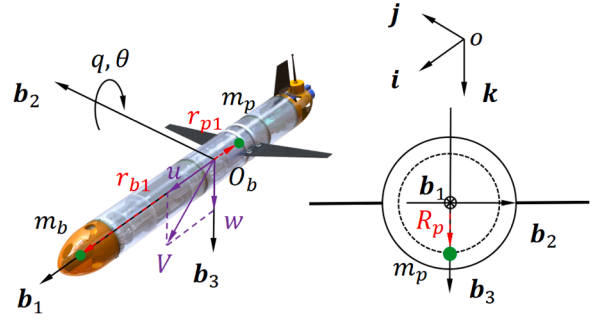


Fig. 1. Schematic of UG's position in the vertical plane.

Table 2

Geometric and hydrodynamic parameters of the UG.

Parameters	Value
Moving mass block	$m_p = 11 \text{ kg}$
Net buoyancy	$-0.5 \text{ kg} \leq m_b \leq 0.5 \text{ kg}$
Slider displacement range	$-0.1 \text{ m} \leq r_{p1} \leq 0.1 \text{ m}$
Displaced fluid mass	$m = 65.28 \text{ kg}$
Slider eccentricity	$R_p = .014 \text{ m}$
Added mass terms	$M_f = \text{diag}(1.48 \text{ kg}, 49.58 \text{ kg}, 65.92 \text{ kg})$
Added inertia terms	$I_f = \text{diag}(0.53 \text{ kg}\cdot\text{m}^2, 7.88 \text{ kg}\cdot\text{m}^2, 10.18 \text{ kg}\cdot\text{m}^2)$
Drag coefficients	$K_{D0} = 7.19$
Lift coefficients	$K_{L0} = -0.36, K_L = 440.99$
Pitch moment coefficients	$K_{M0} = 0.28, K_M = -65.84, K_q = -205.64$

where x and z represent the horizontal and vertical positions in the inertial frame $O - ijk$, respectively. The variables u and w denote the surge and heave velocities in the body-fixed frame $O_b - b_1b_2b_3$, θ is the pitch angle, and q is the pitch angular velocity. The variable r_{p1} denotes the slider position along the b_1 axis in the attitude control mechanism, and m_b is the mass of the external oil bag. Fig. 1 illustrates the UG's position distribution in the vertical plane. The associated geometric and hydrodynamic parameters are provided in Table 2.

The dynamic model in Eq. (11) can be rewritten in a compact form as:

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{x}_2, \\ \dot{\mathbf{x}}_2 = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{U} + \mathbf{d}. \end{cases} \quad (12)$$

where $\mathbf{x} = [x_1, x_2]^T$, $\mathbf{x}_1 = [\theta, z]^T$ is the output vector with θ representing the pitch angle and z the depth of motion; $\mathbf{x}_2 = [\dot{\theta}, \dot{z}]^T$ is the derivative state vector; $\mathbf{U} = [r_{p1}, m_b]^T$ is the control input vector; $\mathbf{d} = [d_1, d_2]^T$ denotes external disturbances; $\mathbf{g}(\mathbf{x}) = \text{diag}\left(-\frac{1}{I_{f2}} m_p g \cos \theta, g \left(\frac{\sin^2 \theta}{M_{f1}} + \frac{\cos^2 \theta}{M_{f3}}\right)\right)$; and $\mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x})]^T$, where $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ expressions are respectively:

$$\begin{aligned} f_1(\mathbf{x}) &= \frac{1}{I_{f2}} \left[(M_{f3} - M_{f1} + K_M)uw - m_b r_{b1} g \cos \theta - m_p g (r_{p1} \cos \theta + R_p \sin \theta) \right. \\ &\quad \left. + K_{M0}u^2 + K_q qu^2 \right], \end{aligned} \quad (13)$$

$$\begin{aligned} f_2(\mathbf{x}) &= -\frac{1}{M_{f1}} [-K_{D0}u^2 + K_{L0}uw] \sin \theta + \frac{1}{M_{f3}} [-K_{L0}u^2 \\ &\quad - (K_L + K_{D0})uw] \cos \theta - uq \cos \theta - wq \sin \theta, \end{aligned} \quad (14)$$

Assumption 1. The matching external time-varying disturbance of the UG system is expressed as $\mathbf{d}$. In Eq. (12), it is assumed to be bounded as $\|\mathbf{d}\| < D$, where $D > 0$ is a known constant.

3. Controller design

This section discusses the fixed-time tracking control strategy of UGs under constrained tasks. The control objectives include two aspects:

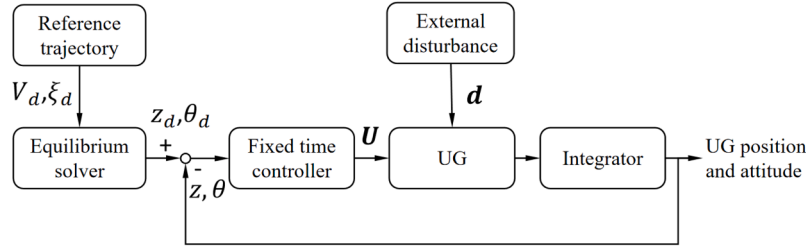


Fig. 2. Fixed-time control block diagram of UG.

(i) Design of a generalizable fixed-time tracking controller for the UG generalized model. The schematic diagram of the control framework is shown in Fig. 2. (ii) Design of a controller for the proposed algorithm on a self-developed UG prototype.

3.1. Fixed-time controller design

The tracking error of system (12) is defined as follows:

$$e_1 = \mathbf{x}_1 - \mathbf{x}_{1d}, \quad (15)$$

where $\mathbf{x}_{1d}$ is the desired trajectory. Differentiating Eq. (15) yields:

$$e_2 = \dot{\mathbf{x}}_1 - \dot{\mathbf{x}}_{1d}, \quad (16)$$

in which $e_2 = \dot{e}_1$, and differentiating Eq. (16), and substituting from system (12), the error dynamics are given by:

$$\begin{cases} \dot{e}_1 = e_2, \\ \dot{e}_2 = f(\mathbf{x}) + g(\mathbf{x})U + d - \ddot{\mathbf{x}}_{1d}. \end{cases} \quad (17)$$

For the UG error system (17), inspired by Liu et al. (2025), the following non-singular terminal sliding surface is designed:

$$s = e_2 + K \text{sig}^{\beta_1}(e_1), \quad (18)$$

where $s = [s_1, s_2]^T$ is the sliding mode variable; $e_1 = [e_{11}, e_{12}]^T$ is the tracking error vector, $e_{11} = \theta - \theta_d$ donates the pitch error, $e_{12} = z - z_d$ donates the depth error, θ_d and z_d are the desired pitch angle and depth respectively; $e_2 = [e_{21}, e_{22}]^T$ is the error derivative vector; $K = \text{diag}(k_1, k_2)$ is the positive definite gain matrix, $k_i > 0$ ($i = 1, 2$); $\text{sig}^{\beta_1}(e_1) = [|e_{11}|^{\beta_1} \text{sign}(e_{11}), |e_{12}|^{\beta_1} \text{sign}(e_{12})]^T$, where $\beta_1 = \frac{m_1+n_1}{2} + \frac{m_1-n_1}{2} \text{sign}(\|e_1\| - 1)$, and $m_1 > 1, 0 < n_1 < 1$.

Remark 1. The design of β_1 provides adaptive convergence behavior based on the magnitude of the tracking error. Specifically, when the error satisfies $\|e_1\| > 1$, $\beta_1 = m_1 > 1$ corresponds to a strong convergence region, where the term $|e_{1i}|^{m_1} \text{sign}(e_{1i})$ ensures rapid convergence of the tracking error e_{1i} toward the origin. Conversely, when $\|e_1\| < 1$, $\beta_1 = n_1 < 1$ defines a precision convergence region, in which the term $|e_{1i}|^{n_1} \text{sign}(e_{1i})$ guarantees that the tracking error converges to the origin within a fixed time.

The proposed controller consists of an equivalent control component and a switching control component. Differentiating Eq. (18) gives:

$$\dot{s} = \dot{e}_2 + \beta_1 K \mathcal{E} e_2, \quad (19)$$

where $\mathcal{E} = \text{diag}(|e_{11}|^{\beta_1-1}, |e_{12}|^{\beta_1-1})$. Substituting Eq. (19), the equivalent control law is formulated as:

$$u_{eq} = f(\mathbf{x}) - \ddot{\mathbf{x}}_{1d} + \beta_1 K \mathcal{E} e_2. \quad (20)$$

When $\|e_1\| < 1$, the exponent $\beta_1 - 1 = n_1 - 1$ becomes negative due to $0 < n_1 < 1$, which may lead to singularities in $\mathcal{E}$. To mitigate this issue,

a saturation function is introduced, as proposed in Feng et al. (2013), to replace the singular term with a bounded approximation:

$$\text{sat}(\mathbf{x}, \epsilon_0) = \begin{cases} \mathbf{x}, & \|\mathbf{x}\| \leq \epsilon_0, \\ \epsilon_0 \text{sgn}(\mathbf{x}), & \|\mathbf{x}\| > \epsilon_0. \end{cases} \quad (21)$$

where ϵ_0 is small constant, $\text{sgn}(\mathbf{x}) = [\text{sgn}(x_1), \text{sgn}(x_2), \dots, \text{sgn}(x_n)]^T$. Accordingly, the equivalent control law is updated as follows:

$$u_{eq} = f(\mathbf{x}) - \ddot{\mathbf{x}}_{1d} + \text{sat}(\beta_1 K \mathcal{E} e_2, \epsilon_0). \quad (22)$$

The switching control law is designed as:

$$u_{sw} = H \text{sig}^{\beta_2}(s) + D \text{sgn}(s), \quad (23)$$

where

$$\text{sig}^{\beta_2}(s) = [|s_1|^{\beta_2} \text{sign}(s_1), |s_2|^{\beta_2} \text{sign}(s_2)]^T, \quad \text{sgn}(s) = [\text{sign}(s_1), \text{sign}(s_2)]^T, \text{ and } H = \text{diag}(h_1, h_2) \text{ is a positive definite diagonal gain matrix with } h_i > 0 \text{ for } i = 1, 2. \text{ The exponent } \beta_2 \text{ is defined as: } \beta_2 = \frac{m_2+n_2}{2} + \frac{m_2-n_2}{2} \text{sign}(\|s\| - 1), \text{ with parameters satisfying } m_2 > 1 \text{ and } 0 < n_2 < 1.$$

To reduce the chattering effect caused by the discontinuous sign function in Eq. (23), the hyperbolic tangent function $\tanh(\frac{s}{\rho_0})$ is adopted as a smooth approximation of $\text{sgn}(s)$, where ρ_0 is a small constant. The resulting switching control law is then given by:

$$u_{sw} = H \text{sig}^{\beta_2}(s) + D \tanh\left(\frac{s}{\rho_0}\right). \quad (24)$$

Finally, by combining the equivalent control law in Eq. (22) with the error dynamics in Eq. (17), the fixed-time sliding mode control (FSMC) input is derived as follows:

$$U = -g^{-1}(\mathbf{x})(u_{eq} + u_{sw}). \quad (25)$$

Theorem 1. Consider the UG system described by Eq. (12). If the control input is designed as in Eqs. (22), (24), and (25), and the control parameters are properly selected, then the system tracking error will converge to a small neighborhood around the origin within a fixed time. Furthermore, the convergence time t_f is independent of the initial conditions and satisfies the upper bound $t_f \leq t_{f1} + t_{f2} + t_{f3}$, where the terms t_{f1} , t_{f2} , and t_{f3} correspond to the convergence phases detailed in the proof.

Proof. Define the Lyapunov candidate function as:

$$V_1 = s^T s, \quad (26)$$

Taking its derivative and expanding yields:

$$\begin{aligned} \dot{V}_1 &= 2s^T \dot{s} \\ &= 2s^T (\dot{e}_2 + \beta_1 K \mathcal{E} e_2) \\ &= 2s^T (f(\mathbf{x}) + g(\mathbf{x})U - \ddot{\mathbf{x}}_{1d} + d + \beta_1 K \mathcal{E} e_2) \\ &= 2s^T (-\text{sat}(\beta_1 K \mathcal{E} e_2, \epsilon_0) + \beta_1 K \mathcal{E} e_2 + d - H \text{sig}^{\beta_2}(s) - D \tanh\left(\frac{s}{\rho_0}\right)). \end{aligned} \quad (27)$$

According to Lemma 1 and Assumption 1, the disturbance-related term satisfies:

$$s^T \left(d - D \tanh\left(\frac{s}{\rho_0}\right) \right) \leq \|s\| \|d\| - D s^T \tanh\left(\frac{s}{\rho_0}\right)$$

$$\begin{aligned}
&= -Ds^T \tanh\left(\frac{s}{\rho_0}\right) + D\|s\| - D\|s\| + \|s\|\|d\| \\
&\leq D\left(\|s\| - s^T \tanh\left(\frac{s}{\rho_0}\right)\right) + \|s\|(-D + \|d\|) \\
&\leq 2 \times 0.2785\rho_0 D.
\end{aligned} \quad (28)$$

In the non-singular region $\|\beta_1 K E e_2\| \leq \varepsilon_0$, the saturation function satisfies:

$$\text{sat}(\beta_1 K E e_2, \varepsilon_0) = \beta_1 K E e_2, \quad (29)$$

Substituting Eqs. (28) and (29) into Eq. (27), we obtain:

$$\begin{aligned}
\dot{V}_1 &\leq -2s^T H \text{sig}^{\beta_2}(s) + 2 \times 0.2785\rho_0 D \\
&\leq -2\lambda_{\min}(H)\|s\|^{\beta_2+1} + 2 \times 0.2785\rho_0 D \\
&\leq -2\varpi V_1^{\frac{\beta_2+1}{2}} + \Psi,
\end{aligned} \quad (30)$$

where $\varpi = \lambda_{\min}(H)$ represents the minimum eigenvalue of the matrix H , and $\Psi = 2 \times 0.2785\rho_0 D$. By Lemma 3, the error tracking will converge to a small neighborhood H_f within a fixed time range t_{f2} , and near the sliding surface $s = 0$, its convergence time satisfies:

$$t_{f2} = \frac{1}{\varpi \bar{\kappa}(m_2 - 1)} + \frac{1}{\varpi \bar{\kappa}(1 - n_2)}, \quad 0 < \bar{\kappa} < 1. \quad (31)$$

In the singular region $\|\beta_1 K E e_2\| > \varepsilon_0$, it follows from Eq. (17) that:

$$e_1(t) = e_1(0) + \int_0^t e_2(\tau) d\tau. \quad (32)$$

The region $S_1 \triangleq \{e_2 : \|\beta_1 K E e_2\| > \varepsilon_0\}$ is defined as the singular region, while $S_2 \triangleq \{e_2 : \|\beta_1 K E e_2\| \leq \varepsilon_0\}$ is defined as the non-singular region.

Within the singular region S_1 , if the i -th component of the error velocity satisfies $e_{2i} > 0$, then the corresponding tracking error e_{1i} increases monotonically. Conversely, when $e_{2i} < 0$, the value of e_{1i} decreases monotonically. Therefore, regardless of the sign of e_{2i} , the error component e_{1i} will monotonically evolve out of the singular region and enter the non-singular region S_2 in finite time. The maximum time required for this transition is denoted by t_{f3} , which is uniformly bounded. For detailed theoretical justification, refer to Feng et al. (2013) and Ni et al. (2016).

In summary, with properly selected control parameters, the convergence neighborhood H_f can be made arbitrarily small. When this neighborhood approaches zero, it can be reasonably approximated that $s = 0$. Substituting this condition into Eq. (18) yields the reduced error dynamics:

$$\dot{e}_1 = -K \text{sig}(e_1). \quad (33)$$

Define the Lyapunov function as:

$$V_2 = e_1^T e_1, \quad (34)$$

then, the Eq. (34) is derived as follows:

$$\begin{aligned}
\dot{V}_2 &= 2e_1^T \dot{e}_1 = -2e_1^T K \text{sig}(e_1) \\
&\leq -2\lambda_{\min}(K)\|e_1\|^{\beta_1+1} \\
&= -2\lambda_{\min}(K)V_2^{\frac{\beta_1+1}{2}},
\end{aligned} \quad (35)$$

where $\lambda_{\min}(K)$ denotes the minimum eigenvalue of matrix K .

According to Lemma 2, e_1 can converge to the origin in a fixed time, and the convergence time satisfies:

$$t_{f1} \leq \frac{1}{\lambda_{\min}(K)(m_1 - 1)} + \frac{1}{\lambda_{\min}(K)(1 - n_1)}. \quad (36)$$

Therefore, the total convergence time of the closed-loop system is bounded by a constant and satisfies:

$$t_f \leq t_{f1} + t_{f2} + t_{f3}. \quad (37)$$

This completes the proof of Theorem 1. $\square$

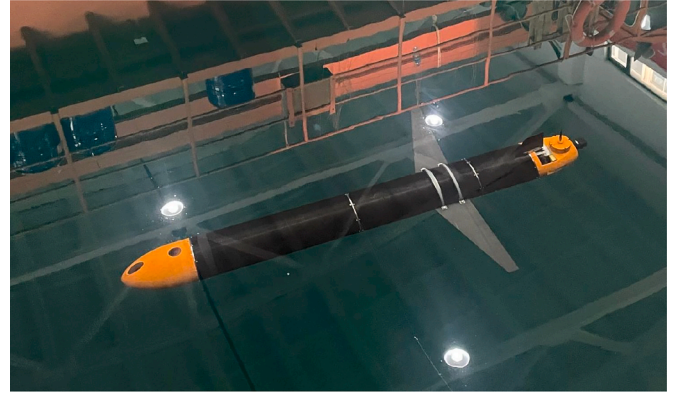


Fig. 3. Independently developed UG prototype.

Remark 2. The fixed-time control algorithm for UG proposed in this paper has faster convergence than the traditional sliding mode control scheme (Yang and Ma, 2010; Zhou et al., 2020). It is independent of the initial conditions compared to the finite-time tracking algorithm (Feng et al., 2002, 2013; Zhang et al., 2023). We propose a novel fixed-time sliding surface and control law, which differs from previous fixed-time control studies for underwater vehicles (Su et al., 2021; Sun et al., 2022; Zhu et al., 2023; Chen et al., 2023). And compared with Liu et al. (2025), this paper generalizes the fixed-time algorithm to the second-order underwater vehicle system and gives a STM32-based controller deployment scheme suitable for UGs.

3.2. UG prototype deployment control method

This section will introduce the UG used in the experimental verification in detail, focusing on the pitch angle tracking task of the UG, and introduce how to deploy the fixed-time control method mentioned in Section 3.1 on the prototype.

The independently developed UG prototype is shown in Fig. 3. UG prototype supports multiple motion modes including vertical sawtooth gliding, large pitch-angle profiling, rudder-based propulsion, and hybrid drive. The system is designed for smooth mode switching and fine-resolution observation in complex mission scenarios. Moreover, this section primarily focuses on verifying its vertical plane gliding capability. A systematic introduction of the UG's mechanical is provided, followed by closed-loop tank experiments that compare the trajectory tracking performance of fixed-time control algorithms.

As illustrated in Fig. 4, the UG adopts a modular compartmental design, comprising four major subsystems: buoyancy adjustment, attitude adjustment, rudder propulsion, and electronic control. The main control board of this paper selects the STM32F427VGT microcontroller based on the ARM Cortex-M4 core architecture. The deployment strategy of the detailed tracking control algorithm for the slider and pitch angle on the UG main control board is given later. This section emphasizes the design and operation of the buoyancy and attitude adjustment systems.

Buoyancy adjustment system: Unlike traditional UG systems that use motor-driven hydraulic pumps for oil exchange (Cao et al., 2019), this prototype employs an electric push rod to actuate a multi-piston parallel hydraulic system (see Fig. 5). The push rod drives four piston cylinders in synchronized reciprocating motion, delivering hydraulic oil to an external variable-volume bladder via a hydraulic manifold. This setup enables continuous adjustment of net buoyancy for ascent and descent control. The system has a maximum discharge volume of approximately 600 mL.

The push rod motor has a self-locking mechanism, which allows the prototype to maintain buoyancy when power is cut off. Table 3 lists the main parameters of the electric actuators. A wire-type displacement sensor is installed between the cylinder base and the piston mounting

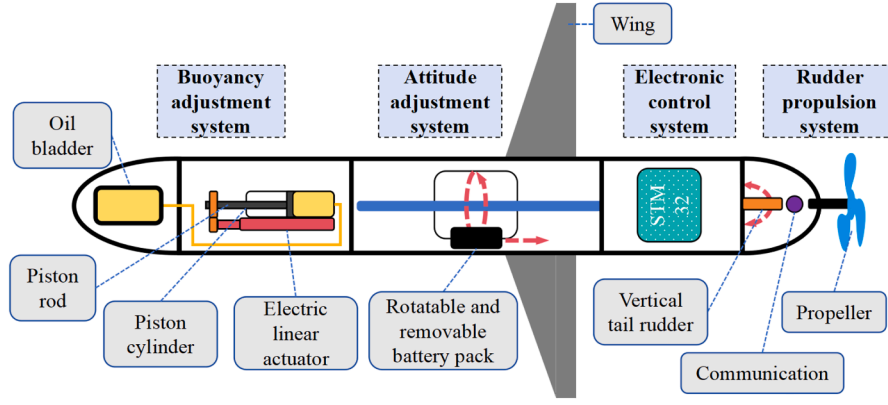


Fig. 4. Main structural layout of the UG prototype.

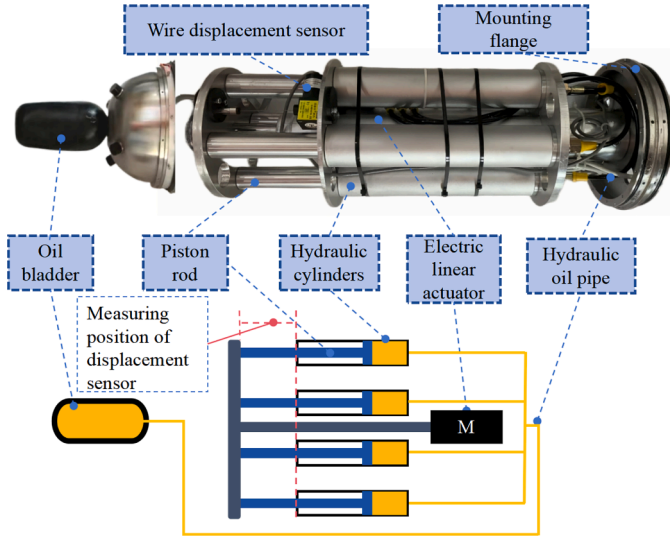


Fig. 5. Buoyancy adjustment mechanism.

plate to measure axial displacement, enabling indirect estimation of oil volume. The safe stroke range of the piston rod is $r_{b1} \in [30, 120]$ mm. When $r_{b1} = 120$ mm, the piston is fully retracted, and buoyancy is minimal. When $r_{b1} = 30$ mm, the oil volume in the external bladder is maximized, yielding maximum buoyancy. Water tank calibration tests show that neutral buoyancy is achieved at $r_{b1} = 57$ mm.

Attitude adjustment system: As shown in Fig. 6, the attitude control module enables both rotational and translational adjustments of a movable slider composed of two 6S lithium battery packs. The rotation mechanism uses a DC motor with a worm-gear reducer to achieve 360° continuous rotation. The translation mechanism drives the slider linearly along the glider's longitudinal axis via a belt-and-ball-screw system. Three high-precision guide rails are incorporated to enhance stability and alignment, and a travel-limit switch is installed at both ends to prevent overrun.

Slider displacement is measured using a rod-type displacement sensor with a resolution of 0.01 mm. The slider's range of motion is $r_{p1} \in [75, 175]$ mm, allowing a maximum travel of 100 mm. When $r_{p1} = 75$ mm, the slider is located near the bow, shifting the center of gravity forward and producing a nose-down pitch; conversely, at $r_{p1} = 175$ mm, the slider is near the stern, resulting in a nose-up configuration. Static tests (see Fig. 7) indicate that the prototype supports an attitude adjustment range of $[-47^\circ, 48^\circ]$. Note that the buoyancy adjustment system can also indirectly affect pitch angle, implying that the glider's full operational pitch range exceeds the values shown here.

Table 3

Motor specifications of UG prototype.

Function	Load	Speed	Voltage (DC)
Electric push rod	5000 N	1.8 mm/s	24 V
Translation motor	5 kg · cm	10 r/min	24 V
Rotation motor	25 kg · cm	12 r/min	12 V

It is worth noting that the displacement sensor in the translation mechanism is installed in reverse, meaning an increase in r_{p1} corresponds to a rearward shift of the center of gravity-opposite to the conventional perception. This arrangement does not affect functionality but should be carefully considered in controller implementation.

In order to enable the full-range rotation of the lithium battery pack, the rotational mechanism employs a worm gear reduction DC motor with a self-locking feature. The motor drives the mass block through a 3:1 gear reduction ratio, achieving 360° continuous rotation. The rotation angle is measured by a Hall-effect angle sensor coupled to the gear shaft, which provides signals that are filtered and processed for feedback in closed-loop control. The specifications of the rotation motor, selected based on load requirements, are listed in Table 3.

The ball screw mechanism used in the translational module features a shaft diameter of 10 mm, a lead of 6 mm, and an overall length of 500 mm. It is supported by essential components including fixed supports, bearing blocks, and nut housings. To enhance rotational precision and structural reliability, the mechanism is supported by a pair of deep groove ball bearings and is mounted on a flange end face. Additionally, to ensure safe cable management during rotation, a spiral spring cable structure is adopted, effectively preventing wire entanglement during continuous motion. Detailed descriptions of the rudder propulsion and electronic control systems are omitted in this section for brevity.

Remark 3. To experimentally validate the feasibility and control performance of the proposed algorithms, this study primarily focuses on in-situ pitch angle regulation as the test scenario. This choice is due to spatial constraints of the pool experiments, which limit the possibility of conducting multi-cycle sawtooth gliding trials.

Under neutral buoyancy conditions (Fig. 3), the glider exhibits dynamic behavior distinct from that observed in standard gliding states. In this configuration, lift-generated torque is negligible, and the dominant torque in pitch arises from hydrodynamic rotational damping. Taking this into account, the general dynamics equation (Eq. (11)) is simplified to a single-degree-of-freedom pitch control model, formulated as follows:

$$\begin{cases} \dot{\theta} = q, \\ \dot{q} = \frac{1}{I_{f2}} [-m_p r_{p1} \dot{r}_{p1} q - m_p g (r_{p1} \cos \theta + R_p \sin \theta) + K_q q]. \end{cases} \quad (38)$$

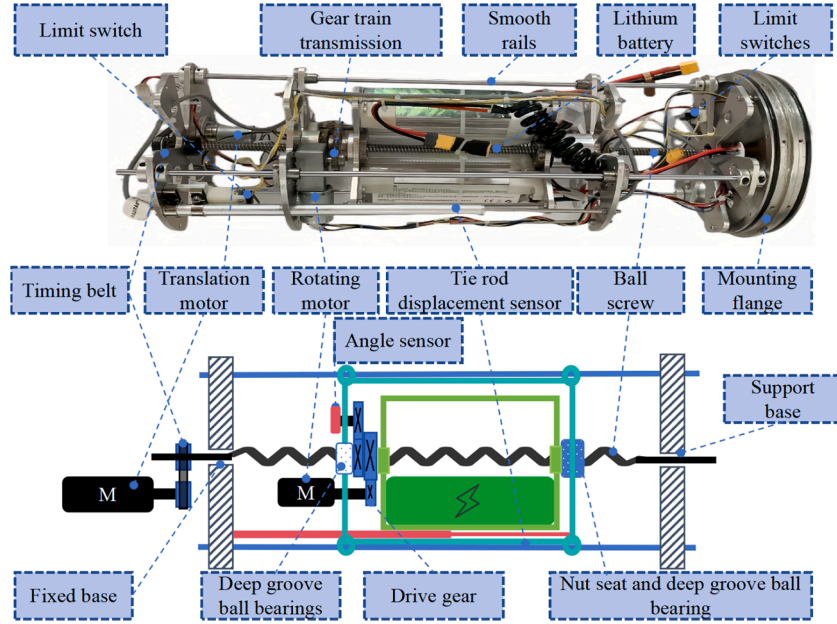


Fig. 6. Attitude adjustment mechanism.

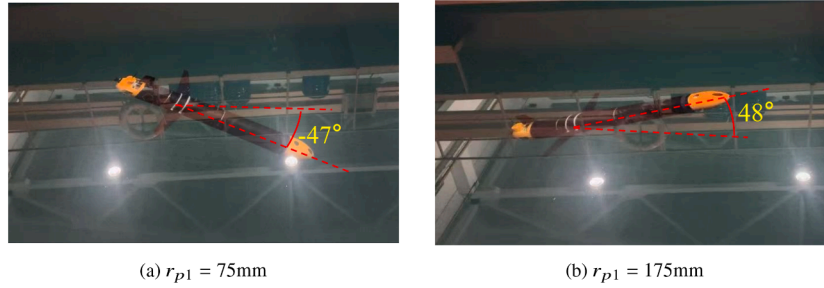


Fig. 7. Pitch angle testing of attitude adjustment system.

In this model, r_{p1} and its time derivative $\dot{r}_{p1}$ serve as system inputs, while the pitch angle θ and pitch rate q are the system outputs. The key modeling parameters obtained through parameter identification are: pitch moment of inertia $I_{f2} = 23.09 \text{ kg}\cdot\text{m}^2$, offset distance of the slider mass center $R_p = .0488 \text{ m}$, and rotational damping coefficient $K_q = -2.14 \text{ kg}\cdot\text{s}/\text{rad}^2$. With the known slider mass $m_p = 4.6 \text{ kg}$ and gravitational acceleration $g = 9.8 \text{ m}/\text{s}^2$, a complete dynamic model of the in-situ pitch control system is constructed for experimental verification and controller design.

The outer-loop system is reformulated into the following state-space representation:

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = f(x) + g(x)u_{r_{p1}}. \end{cases} \quad (39)$$

where $x_1 = \theta$, $x_2 = q$ represent the system states, and the nonlinear term is $f(x) = -\frac{m_p g R_p}{I_{f2}} \sin \theta + \frac{K_q}{I_{f2}} q$; the control gain is $g(x) = -\frac{m_p}{I_{f2}} (\dot{r}_{p1} q + g \cos \theta)$; the system input is $u_{r_{p1}}$, which also represents the linear position r_{p1} of the slider; the system output is $y = \theta$.

To facilitate control law design, the pitch angle tracking error is defined as:

$$e_p = x_1 - x_{1d} = \theta - \theta_d. \quad (40)$$

where $e_p = \tilde{\theta}$.

The control system is deployed on a translational slider-driven mechanism. Due to mechanical hysteresis and nonlinear friction effects introduced by the synchronous belt and ball screw in the actuation chain,

a nonlinear and complex mapping exists between the PWM duty cycle u_{pwm} and the resulting slider displacement r_{p1} . To characterize this dynamic behavior, the following second-order state-space model is adopted:

$$\begin{cases} \dot{r}_{p1} = v_p, \\ \dot{v}_p = -\frac{1}{\tau_p} v_p + \frac{k_p}{\tau_p} u_{\text{pwm}}. \end{cases} \quad (41)$$

where v_p denotes the linear velocity of the slider. The system parameters are obtained via experimental system identification as $k_p = .0086$ and $\tau_p = 1.54$.

To comply with hardware limitations and extend actuator lifespan, the control input u_{pwm} is subject to the following nonlinear saturation and dead-zone constraints:

$$u_{\text{pwm}} = \begin{cases} 100, & u_{\text{pwm}} \geq 100 \text{ (positive saturation)}, \\ -100, & u_{\text{pwm}} \leq -100 \text{ (negative saturation)}, \\ 0, & -25 < u_{\text{pwm}} < 25 \text{ (dead zone)}, \\ u_{\text{pwm}}, & \text{other.} \end{cases} \quad (42)$$

This constraint design helps mitigate actuator instability caused by frequent switching or saturation. However, it may also introduce performance degradation in closed-loop control.

Given the system characteristics, a hierarchical control architecture is implemented to experimentally validate pitch attitude control on the UG prototype, comprising two control layers:

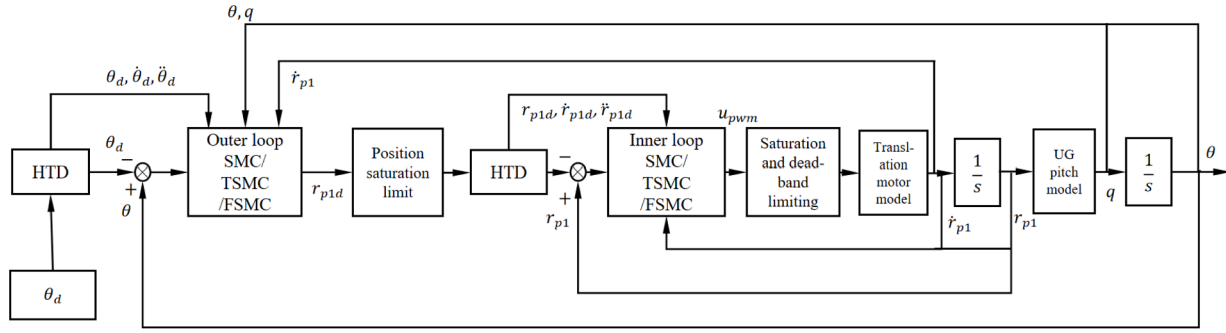


Fig. 8. Schematic diagram of pitch angle control under neutral suspension state.

- Inner loop position control: Utilizes the PWM duty cycle u_{pwm} as the control input to accurately track the reference translational slider position r_{p1d} .
- Outer loop attitude control: Treats r_{p1} as the control input to regulate the pitch angle θ , ensuring attitude stabilization.

The pitch angle tracking control block diagram of the UG prototype in the natural suspension state is shown in Fig. 8.

Inner loop: The tracking error of the moving slider is defined as:

$$e = r_{p1} - r_{p1d}, \quad (43)$$

where r_{p1d} is the reference slider displacement, which is usually a smooth, differentiable and bounded reference trajectory.

The fixed time control law for slider tracking is similar to that in Section 3.1. The sliding surface is designed as

$$s = \dot{e} + k_{f1}|e|^{\beta_{f1}}\text{sign}(e), \quad (44)$$

where the power exponential dynamic adjustment parameter is $\beta_{f1} = \frac{m_{f1}+n_{f1}}{2} + \frac{m_{f1}-n_{f1}}{2}\text{sign}(|e|-1)$, $m_{f1} > 1, 0 < n_{f1} < 1$. The equivalent and switching control laws are

$$u_{xeq} = -\frac{1}{\mathcal{T}_p} \dot{r}_{p1} - \ddot{r}_{p1d} + \text{sat}\left(k_{f1}\beta_{f1}|e|^{\beta_{f1}-1}\dot{e}, \sigma_3\right), \quad (45)$$

$$u_{xsw} = k_{f2}|s|^{\beta_{f2}}\text{sign}(s) + k_{f3} \tanh\left(\frac{s}{\sigma_4}\right), \quad (46)$$

where σ_3 is a positive constant, and the definition of $\text{sat}(\cdot)$ in Eq. (45) is similar to that in Eq. (21). $\beta_{f2} = \frac{m_{f2}+n_{f2}}{2} + \frac{m_{f2}-n_{f2}}{2}\text{sign}(|s|-1)$, $m_{f2} > 1, 0 < n_{f2} < 1$. $\sigma_4 > 0$. Then, the control law is:

$$u_{pwm} = -\frac{k_p}{\mathcal{T}_p} (u_{xeq} + u_{xsw}). \quad (47)$$

Outer loop: The sliding surface of the outer closed loop is defined as:

$$s_p = \dot{e}_p + \bar{k}_{f1}|e_p|^{\bar{\beta}_{f1}}\text{sign}(e_p), \quad (48)$$

where $\bar{k}_{f1}$ is the selected sliding surface parameter. $\bar{\beta}_{f1} = \frac{\bar{m}_{f1}+\bar{n}_{f1}}{2} + \frac{\bar{m}_{f1}-\bar{n}_{f1}}{2}\text{sign}(|e_p|-1)$, $\bar{m}_{f1} > 1, 0 < \bar{n}_{f1} < 1$.

The equivalent control law is:

$$\bar{u}_{xeq} = f(x) - \ddot{x}_{1d} + \text{sat}\left(\bar{k}_{f1}\bar{\beta}_{f1}|e_p|^{\bar{\beta}_{f1}-1}\dot{e}_p, \bar{\sigma}_3\right), \quad (49)$$

where $\bar{\sigma}_3 > 0$, and the switching control law is designed as:

$$\bar{u}_{xsw} = \bar{k}_{f2}|s_p|^{\bar{\beta}_{f2}}\text{sign}(s_p) + \bar{k}_{f3} \tanh\left(\frac{s_p}{\bar{\sigma}_4}\right), \quad (50)$$

where $\bar{\beta}_{f2} = \frac{\bar{m}_{f2}+\bar{n}_{f2}}{2} + \frac{\bar{m}_{f2}-\bar{n}_{f2}}{2}\text{sign}(|s_p|-1)$, $\bar{m}_{f2} > 1, 0 < \bar{n}_{f2} < 1$, $\bar{\sigma}_4 > 0$, $\bar{k}_{f3} > 0$.

Finally, the control law of the fixed-time control scheme for UG prototype pitch angle tracking is obtained as follows:

$$u_{r_{p1}} = -g^{-1}(x)(\bar{u}_{xeq} + \bar{u}_{xsw}). \quad (51)$$

where $u_{r_{p1}} = r_{p1d}$, which is the reference slider position to be tracked in the inner loop.

Remark 4. The parameter selection method of the inner and outer dual-loop FSMC scheme is basically consistent with the inner loop parameter selection method, see Section 3.1 for details.

To improve trajectory smoothness and ensure boundedness and continuity of its derivatives, a high-order tracking differentiator (HTD) is introduced. The HTD serves both as a differentiator and a high-frequency noise suppressor. The fourth-order tracking differentiator is defined as:

$$\begin{cases} \dot{\zeta}_1 = \zeta_2, \\ \dot{\zeta}_2 = \zeta_3, \\ \dot{\zeta}_3 = \zeta_4, \\ \dot{\zeta}_4 = -Y_h^4(\zeta_1 - v_h(t)) - 4Y_h^3\zeta_2 - 6Y_h^2\zeta_3 - 4Y_h\zeta_4. \end{cases} \quad (52)$$

where ζ_1 tracks the reference signal $v_h(t)$. ζ_2 , ζ_3 , and ζ_4 represent its first to third-order derivatives, respectively. The tuning parameter $Y_h > 0$ governs the balance between tracking speed and filtering effectiveness: larger values improve responsiveness but reduce noise attenuation. The optimal choice of Y_h is empirically determined based on experimental performance.

Remark 5. During the deployment of the FSMC scheme on the UG main control board, the second-order derivatives of θ and r_{p1d} in Eqs. (45) and (49) can be obtained using the HTD method.

Remark 6. This paper proposes a fixed-time pitch angle tracking control algorithm for UG, which can be readily implemented on the STM32 main control board. By simplifying the deployment of the control algorithm on the UG prototype, the control parameters become easier to tune and adapt. Moreover, the proposed algorithm is portable and can be extended to the trajectory tracking of other underwater vehicle prototypes, facilitating further verification of its effectiveness.

4. Simulation and experiment

This section evaluates the performance of the proposed control strategy through numerical simulations and laboratory pool experiments. In order to verify the fixed time convergence characteristics of the proposed algorithm, the proposed algorithm is compared with two control schemes, asymptotic convergence and finite-time control, for the pitch angle and position in the vertical plane through numerical simulation. In addition, the proposed algorithm is compared with asymptotic convergence and finite-time control for the pitch angle tracking of the UG prototype in the laboratory pool experiments.

4.1. Simulation of control strategies

To highlight the advantages of the proposed fixed-time sliding mode control (FSMC) strategy, four control approaches are compared using

Table 4
Parameter settings of four control strategies in **Case 1**.

Control Strategy	Parameter Settings
FSMC	$k_1 = 0.15, k_2 = 0.1, m_1 = 1.2, n_1 = 0.8, m_2 = 1.2, n_2 = 0.8, h_1 = 0.001, h_2 = 0.001, D = 0.004, \rho_0 = 0.05, \varepsilon_0 = 0.05$.
SMC	$\kappa_1 = 0.05, \kappa_2 = 0.05, \rho_{11} = 0.5, \rho_{12} = 0.5, \rho_{21} = 0.005, \rho_{22} = 0.005$.
NTSMC	$\Lambda_1 = 0.15, \Lambda_2 = 0.15, \alpha = 0.6, \epsilon = 0.02, l_d = 0.002, \Psi = 0.0002$.
TSMC	$\gamma_1 = 0.02, \gamma_2 = 0.02, p = 5, q = 3, \chi_1 = 0.005, \chi_2 = 0.005$.

the system model (12) and its error dynamics (20), including: conventional sliding mode control (SMC), terminal sliding mode control (TSMC), non-singular terminal sliding mode control (NTSMC), and the proposed FSMC.

1) Sliding mode control (SMC): The sliding surface is defined as:

$$s_c = \kappa e_1 + e_2, \quad (53)$$

where $e_2 = \dot{e}_1$, $s_c = [s_{c1}, s_{c2}]^T$, and $\kappa = \text{diag}(\kappa_1, \kappa_2)$ is a positive definite gain matrix. The control law is given by:

$$u_c = g(x)^{-1} [\ddot{x}_{1d} - f(x) - \kappa^{-1}(\dot{x}_1 - \dot{x}_{1d}) - \rho_1 s_c - \rho_2 \text{sgn}(s_c)], \quad (54)$$

where $\rho_1 = \text{diag}(\rho_{11}, \rho_{12})$ and $\rho_2 = \text{diag}(\rho_{21}, \rho_{22})$ are diagonal control gain matrices respectively.

2) Non-singular terminal sliding mode control (NTSMC): Following the approach in Feng et al. (2013), the sliding surface is defined as:

$$s_n = e_2 + \Lambda \text{sig}^\alpha(e_1), \quad (55)$$

where $\text{sig}^\alpha(e_1) = [|e_{11}|^\alpha \text{sign}(e_{11}), |e_{12}|^\alpha \text{sign}(e_{12})]^T$, $0 < \alpha < 1$; $\Lambda = \text{diag}(\Lambda_1, \Lambda_2)$, $\Lambda_i > 0$ is the positive constant.

The control law is:

$$u_n = g(x)^{-1} (-f(x) + \text{sat}(-\alpha \Lambda N e_2, \epsilon) - Y \text{sgn}(s_n) + \ddot{x}_{1d}), \quad (56)$$

where $\text{sat}(-\alpha \Lambda N e_2, \epsilon)$ denotes the saturation function with boundary $\epsilon > 0$; $N = \text{diag}(|e_{11}|^{\alpha-1}, |e_{12}|^{\alpha-1})$; $Y = l_d + \Psi$ with $\|d\| \leq l_d$ and $\Psi > 0$.

3) Terminal sliding mode control (TSMC): Referring to Feng et al. (2002), the TSMC sliding surface is designed as:

$$s_t = e_1 + \frac{1}{\gamma} \text{sig}^{\frac{p_t}{q_t}}(e_2), \quad (57)$$

where $\text{sig}^{\frac{p_t}{q_t}}(e_2) = [|e_{21}|^{\frac{p_t}{q_t}} \text{sign}(e_{21}), |e_{22}|^{\frac{p_t}{q_t}} \text{sign}(e_{22})]^T$, $1 < \frac{p_t}{q_t} < 2$; $\gamma = \text{diag}(\gamma_1, \gamma_2)$, $\gamma_1 > 0, \gamma_2 > 0$. The corresponding control law is:

$$u_t = \frac{1}{g(x)} \left[-f(x) + \ddot{x}_{1d} - \frac{q_t}{p_t} \gamma \text{sig}^{2-\frac{p_t}{q_t}}(e_2) - \chi \text{sgn}(s_t) \right], \quad (58)$$

in which $\text{sig}^{2-\frac{p_t}{q_t}}(e_2) = [|e_{21}|^{2-\frac{p_t}{q_t}} \text{sign}(e_{21}), |e_{22}|^{2-\frac{p_t}{q_t}} \text{sign}(e_{22})]^T$;

$\chi = \text{diag}(\chi_1, \chi_2)$, $\chi_1 > 0, \chi_2 > 0$; $\text{sgn}(s_t)$ in the control law is a vector of element-by-element sign functions.

Case 1: the simulation compares the trajectory tracking performance of UG in the vertical plane using four control algorithms.

An equilibrium state solver (Zhou et al., 2020) is utilized to convert the desired glide velocity and glide angle into the corresponding reference pitch angle θ_d and depth z_d . The desired glide angle ξ_d and velocity V_d are defined as:

$$\xi_d = \begin{cases} 0.65 \text{ rad}, & \text{if } t \in [0, 100] \cup [200, 300] \cup [400, 500] \cup [600, 700] \text{ s}, \\ -0.65 \text{ rad}, & \text{if } t \in [100, 200] \cup [300, 400] \cup [500, 600] \cup [700, 800] \text{ s}. \end{cases} \quad (59)$$

$$V_d = 0.35 \text{ m/s}. \quad (60)$$

The initial condition is set as $\theta(0) = 0.2 \text{ rad}$ and $z(0) = -0.1 \text{ m}$, while the time-varying external disturbances are defined as $d_1 = 0.001 I_{f2} \sin(5\pi t)$ and $d_2 = 0.001 M_{f1} \sin(5\pi t)$. All controllers adopt appropriately parameters (see Table 4) to ensure a fair performance comparison.

Fig. 9 compares the tracking performance of pitch angle and depth for the four controllers. All schemes achieve acceptable trajectory tracking (Fig. 9(b) and (d)). The conventional SMC exhibits the slowest convergence. FSMC achieves the fastest convergence in pitch tracking, while its advantage in depth tracking is modest. Fig. 10 shows the control inputs. NTSMC and TSMC introduce pronounced chattering in r_{p1} and m_b during steady-state ascent/descent, potentially increasing energy consumption. By contrast, FSMC produces larger transient inputs during mode transitions but maintains smoother steady-state inputs, which favors energy-efficient long-term operation. Overall, FSMC demonstrates better convergence speed and robustness.

Case 2: compare the fixed time convergence performance of the proposed algorithms.

To validate the independence of the FSMC controller's convergence behavior from initial conditions, a comparison is made with TSMC under four distinct initial states:

- **State 1:** $\theta_1 = 0.25 \text{ rad}$, $z_1 = -0.2 \text{ m}$,
- **State 2:** $\theta_2 = 0.2 \text{ rad}$, $z_2 = -0.15 \text{ m}$,
- **State 3:** $\theta_3 = 0.15 \text{ rad}$, $z_3 = -0.1 \text{ m}$,
- **State 4:** $\theta_4 = 0.13 \text{ rad}$, $z_4 = -0.05 \text{ m}$.

The reference glide angle and velocity are set to $\xi_d = 0.6 \text{ rad}$ and $V_d = 0.35 \text{ m/s}$, respectively. Controller parameters remain consistent with Case 1.

Figs. 11 and 12 illustrate the tracking results under the four initial states. TSMC exhibits significant sensitivity to initial conditions, with pitch error convergence times varying between 9 s and 17 s. Depth error convergence similarly fluctuates.

In contrast, FSMC achieves consistent performance: under different initial system states, the pitch angle error converges at about 8 s, and the depth error converges at about 12 s. The simulation comparison of the performance of the two control schemes under different initial conditions is shown in Table 5. These results verify that FSMC possesses fixed-time convergence properties, independent of initial conditions, with an explicitly bounded convergence time. This robustness will be further confirmed through experimental validation.

4.2. Laboratory pool experiments

The pool experiment of UG prototype is mainly divided into the slider tracking of the inner ring and the in-situ pitch angle tracking of the double rings in the suspended state.

Case 3: comparative experiment on slider position tracking performance.

To evaluate the convergence characteristics of the proposed control schemes, an experimental comparison was conducted under consistent initial conditions. The prototype was initially placed in a stationary, balanced state, with the desired slider position set to $r_{p1d} = 90 \text{ mm}$ and the initial position set to $r_{p1}(0) = 120 \text{ mm}$. This offset was intentionally introduced to clearly demonstrate the transient tracking behaviors of different controllers.

Three control algorithms (SMC, TSMC, and FSMC) were implemented in the slider control loop under identical conditions. The control parameters used in each scheme are listed in Table 6. To facilitate the display of the selected parameters of the comparison method, the two algorithms (SMC, TSMC) are briefly introduced.

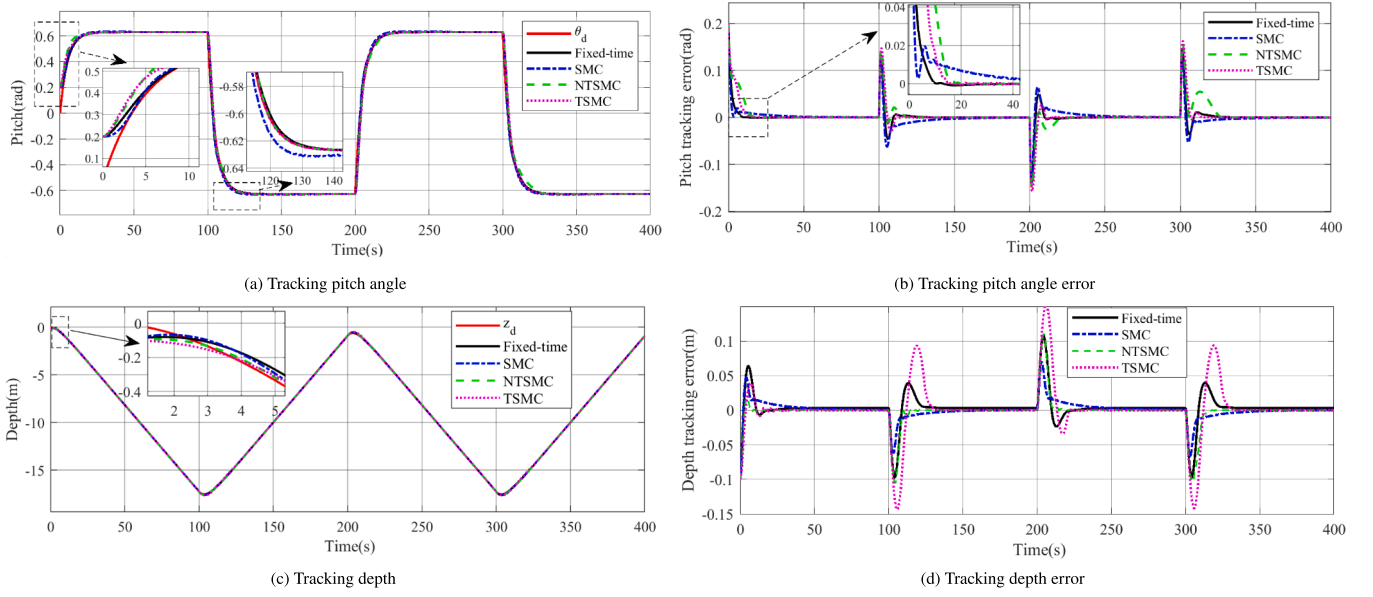


Fig. 9. Comparative analysis of trajectory tracking performance using different control strategies.

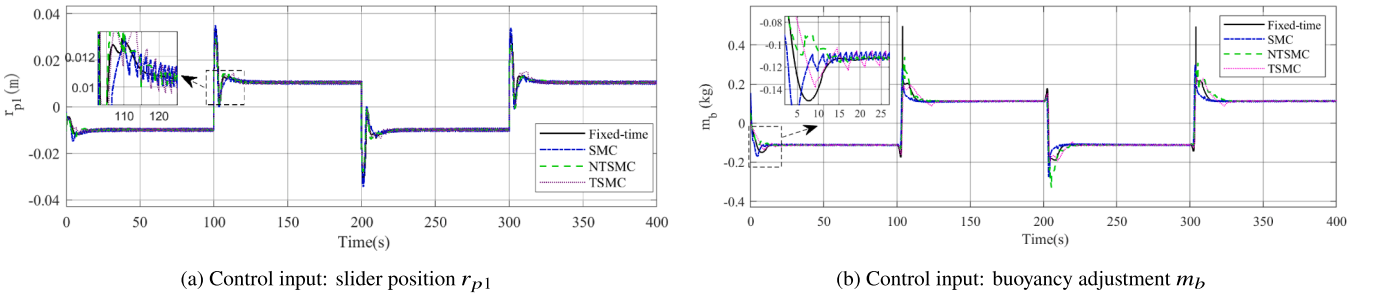


Fig. 10. Comparison of control inputs under different control schemes.

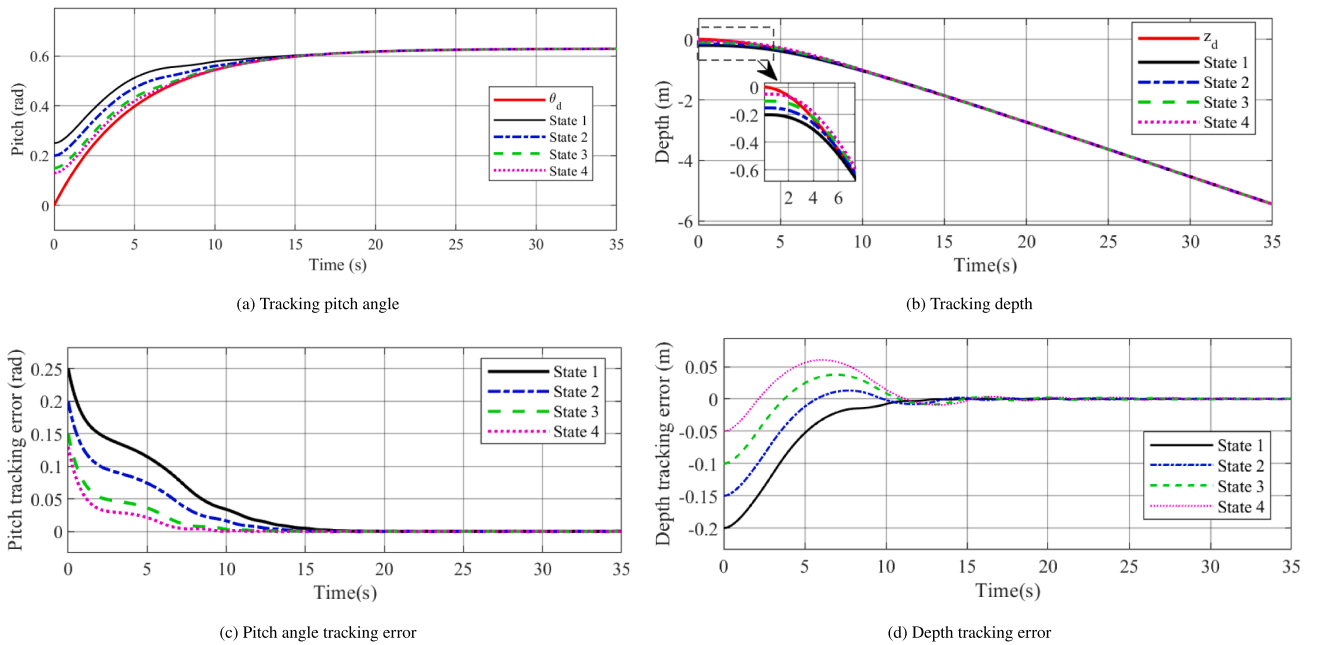


Fig. 11. Tracking performance of the TSMC controller under different initial conditions.

Table 5

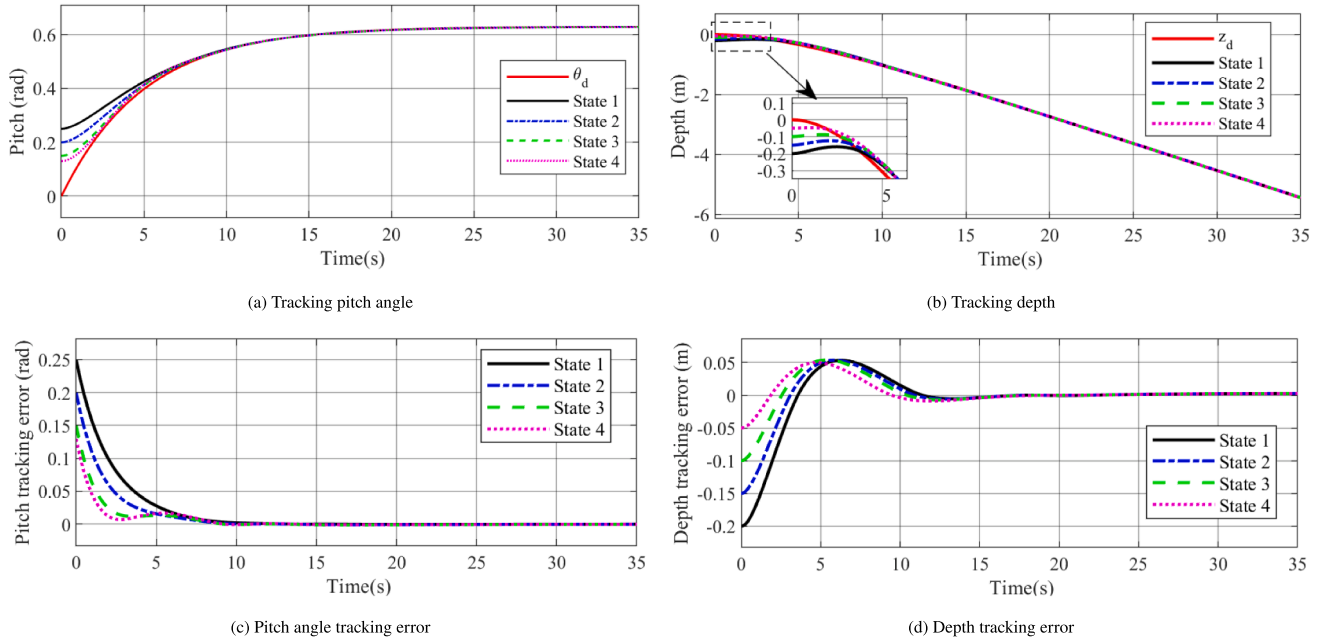
The simulation compares the performance of the two control schemes under different initial conditions in **Case 2**.

Tracking Trajectory	TSMC (Convergence time(s))				FSMC (Convergence time(s))			
	State 1	State 2	State 3	State 4	State 1	State 2	State 3	State 4
Pitch Angle	17	14	12	9	8	8	8	9
Depth	14	16	13	17	11	12	12	13

Table 6

Experimental control parameters for three single-loop control algorithms in **Case 3** and **Case 4**.

Control Scheme	Parameter Values
SMC	$k_{c1} = 0.5, k_{c2} = 0.5, \sigma_1 = 0.2$.
TSMC	$k_{t1} = 0.09, k_{t2} = 0.09, \sigma_2 = 0.04$.
FSMC	$k_{f1} = 0.2, k_{f2} = 0.038, k_{f3} = 0.8, \sigma_3 = 0.1, \sigma_4 = 0.1, m_{f1} = 1.2, n_{f1} = 0.8, m_{f2} = 1.2, n_{f2} = 0.8$.

**Fig. 12.** Tracking performance of the FSMC controller under different initial conditions.

SMC: For the slider tracking error in Eq. (43), define the sliding surface:

$$s = k_{c1}e + \dot{e}, \quad (61)$$

where $k_{c1} > 0$, and the control law is:

$$u_{pwm} = \frac{\mathcal{T}_p}{k_p} \left[\ddot{r}_{p1d} + \frac{1}{\mathcal{T}_p} \dot{r}_{p1} - k_{c1}(\dot{r}_{p1} - \dot{r}_{p1d}) - k_{c2} \text{sat}\left(\frac{s}{\sigma_1}\right) \right], \quad (62)$$

where

$$\text{sat}\left(\frac{s}{\sigma_1}\right) = \begin{cases} \text{sign}(s), & |s| > \sigma_1, \\ \frac{s}{\sigma_1}, & |s| \leq \sigma_1. \end{cases} \quad (63)$$

TSMC: According to the slider tracking error in Eq. (43), define the sliding surface:

$$s = e + \frac{1}{k_{t1}} |\dot{e}|^{\frac{p_t}{q_t}} \text{sign}(\dot{e}), \quad (64)$$

the control law is:

$$u_{pwm} = \frac{k_p}{\mathcal{T}_p} \left[\frac{1}{\mathcal{T}_p} \dot{r}_{p1} + \ddot{r}_{p1d} - \frac{k_{t1} q_t}{p_t} e^{2 - \frac{p_t}{q_t}} - k_{t2} \text{sat}\left(\frac{s}{\sigma_2}\right) \right]. \quad (65)$$

Remark 7. The experimental methods in this section correspond to SMC, TSMC, and FSMC in the Section 4.1.

These parameters were carefully tuned to minimize tracking error while satisfying convergence conditions, taking into account motor dead-zone characteristics and the influence of frequent commutation on the drive electronics. To quantitatively assess the saturation behavior of the control inputs, a saturation index δ_u is defined as:

$$\delta_u = \begin{cases} 100 - u_{pwm}, & \text{if } u_{pwm} > 100, \\ -100 - u_{pwm}, & \text{if } u_{pwm} < -100, \\ 0, & \text{if } -100 \leq u_{pwm} \leq 100. \end{cases} \quad (66)$$

Let $\tilde{r}_{p1} = e$ denote the position tracking error, and u_{pwm} denote the PWM control input to the slider motor.

From Fig. 13(a) and (e), it is evident that all three control schemes are capable of completing the slider tracking task. However, both TSMC and FSMC exhibit significantly faster convergence compared to the conventional SMC. Specifically, SMC requires approximately 80 s to reach convergence, whereas TSMC and FSMC converge within approximately 50 s—a reduction of 30 s in response time. As illustrated in Fig. 13(d), the FSMC scheme activates the saturation control component primarily during the initial large-error phase. Once the tracking error falls below the designated threshold, the control input stabilizes rapidly, and actuator engagement becomes intermittent and minimal. This behavior indicates that FSMC achieves a desirable balance between rapid response and control input efficiency, effectively suppressing small fluctuations caused by

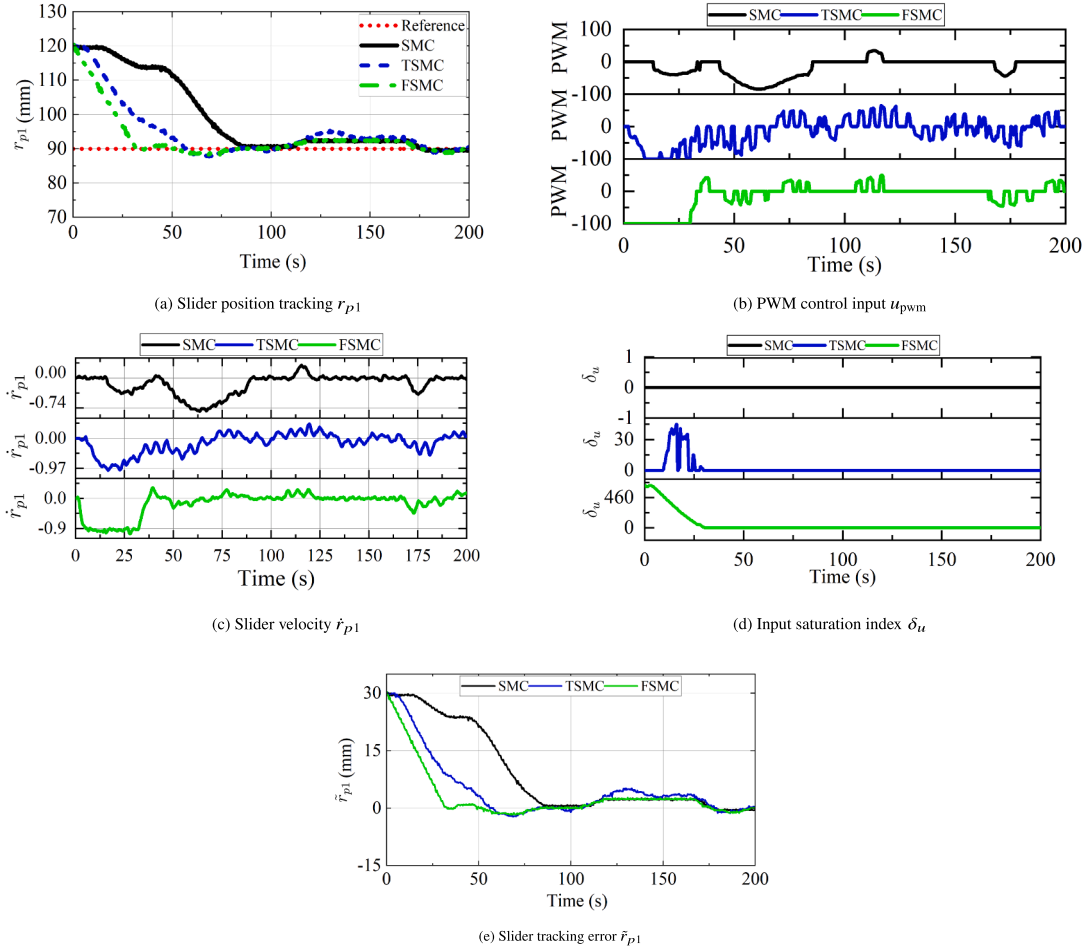


Fig. 13. Comparison of slider tracking performance under different control schemes with an initial position of $r_{p1}(0) = 120$ mm.

sensor noise or drift. In contrast, while the TSMC controller also demonstrates fast convergence, it exhibits minor residual oscillations in the actuator after reaching the target position. These fluctuations are likely induced by frequent switching near the motor's dead-zone threshold and may pose limitations in energy-sensitive applications or scenarios requiring minimal actuator wear.

Case 4: sensitivity analysis for different initial conditions of slider position tracking.

To further evaluate the robustness of each control scheme to varying initial conditions, four initial slider positions were selected: $r_{p1}(0) = 95$ mm, 100 mm, 110 mm, and 120 mm. In all cases, the desired position was maintained at $r_{p1d} = 90$ mm. The control parameters remained consistent with those listed in Table 6.

By comparing the convergence time, overshoot behavior, and actuator saturation characteristics across these scenarios, the sensitivity of each control strategy to initial condition variations is assessed.

Fig. 14 presents the experimental results of slider position tracking under varying initial conditions for three control strategies. The results demonstrate that as the initial position deviates further from the target, the convergence time of both SMC and TSMC increases significantly. Specifically, as shown in Fig. 14(a), the convergence time of the SMC controller increases from 50 s to 90 s as the initial state moves away from the equilibrium point. Similarly, Fig. 14(b) illustrates that the convergence time of TSMC grows from 25 s to 52 s. In contrast, the FSMC scheme maintains a nearly constant convergence time of approximately 28 s across all initial conditions, thereby validating its fixed-time convergence property. These findings further confirm that the proposed FSMC

algorithm exhibits initial-state-independent convergence and superior robustness, which is particularly advantageous in practical scenarios with unpredictable disturbances or large initial deviations.

The results from Case 3 and Case 4 collectively verify that all three control schemes-asymptotic (SMC), finite-time (TSMC), and fixed-time (FSMC)-are capable of achieving accurate slider tracking in the translational motor-slider system. Among them, the FSMC controller demonstrates the best overall performance due to its fast convergence, insensitivity to initial conditions, and effective suppression of input chattering. These advantages highlight its strong potential for engineering applications requiring high stability and responsiveness under dynamic and uncertain operating environments.

Case 5: performance analysis of pitch angle tracking under varying initial conditions in the pool environment.

This experiment aims to evaluate the effect of different control strategies on the pitch angle tracking performance of UG prototype under a neutral suspension state. The initial test condition is set as $r_{p1}(0) = 125$ mm, corresponding to an initial pitch angle of $\theta(0) \approx -2^\circ$, which places the UG prototype in a suspended state. Considering the physical characteristics of the system, a hierarchical control structure is adopted. The inner loop, consistent with that used in Case 3 and Case 4, is responsible for accurate tracking of the slider position. The outer loop, designed based on the pitch dynamics model in Eq. (38), focuses on comparing the performance of three control strategies: traditional sliding mode control (SMC), terminal sliding mode control (TSMC), and FSMC (Control method see Eqs. (44)-(51)). The overall control framework is illustrated in Fig. 8.

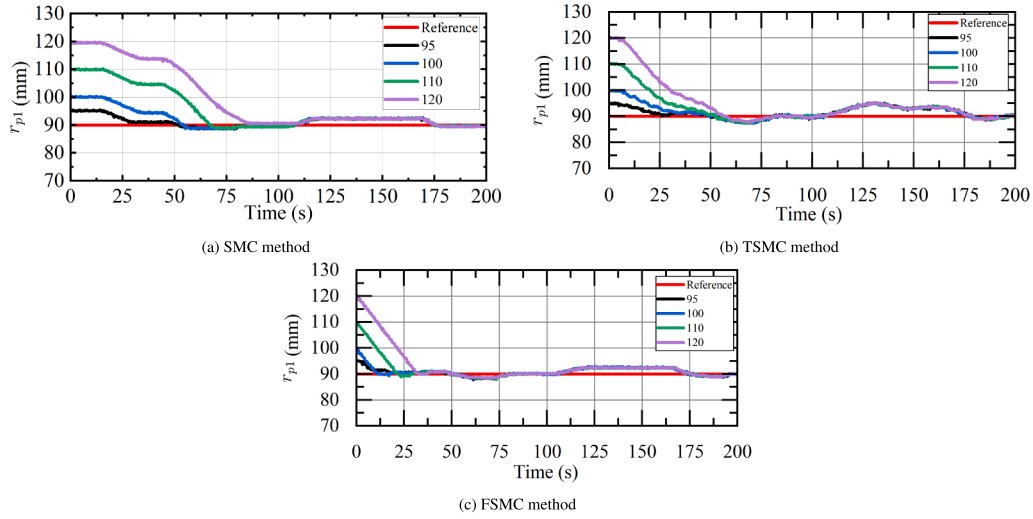


Fig. 14. Comparison of slider tracking performance under varying initial conditions for single-loop three control schemes.

Table 7

Experimental parameters for three control schemes under neutral suspension condition in Case 5.

Control Scheme	Parameter Values
SMC	$k_{c1} = 0.25, k_{c2} = 0.4, \sigma_1 = 0.08, \bar{k}_{c1} = 0.2, \bar{k}_{c2} = 0.4, \bar{\sigma}_1 = 0.15.$
TSMC	$k_{r1} = 0.04, k_{r2} = 0.1, \sigma_2 = 0.2, \bar{k}_{r1} = 0.05, \bar{k}_{r2} = 0.1, \bar{\sigma}_2 = 0.01.$
FSMC	$k_{f1} = 0.15, k_{f2} = 0.1, k_{f3} = 0.05, \sigma_3 = 0.2, \sigma_4 = 0.1, m_{f1} = 1.2, n_{f1} = 0.8, m_{f2} = 1.2, n_{f2} = 0.8,$ $\bar{k}_{f1} = 0.05, \bar{k}_{f2} = 0.08, \bar{k}_{f3} = 0.01, \bar{\sigma}_3 = 0.1, \bar{\sigma}_4 = 0.02, \bar{m}_{f1} = 1.2, \bar{n}_{f1} = 0.8, \bar{m}_{f2} = 1.2, \bar{n}_{f2} = 0.8.$

The structure of the outer-loop controllers for SMC and TSMC follows the design methodology adopted in the inner loop. The corresponding parameter values used in the experiments are listed in Table 7. This mainly involves the change of parameter variables. The other two control schemes both add the symbol $\bar{\cdot}$ to the parameter value to represent the outer loop design parameters.

As illustrated in Fig. 8, to ensure the smoothness and boundedness of the desired pitch trajectory and its first- and second-order derivatives, a HTD method is employed in the outer loop control. The pitch angle reference signal is smoothed using the HTD method with the tuning parameter $Y_h = 0.2$. Simultaneously, to suppress high-frequency noise induced by the slider position sensor, the target slider position r_{pl} , calculated by the outer loop, is filtered using an HTD with $Y_h = 1.5$. Moreover, the nonlinear characteristics of the translational motor, including input saturation and dead-zone effects, are taken into account in the control implementation. The parameter settings for the dual-loop SMC, TSMC, and FSMC schemes are summarized in Table 7.

To evaluate the robustness of the control algorithms against variations in initial conditions, three initial pitch angles were selected: -8° ($r_{pl}(0) = 120\text{mm}$), 5° ($r_{pl}(0) = 130\text{mm}$), 10° ($r_{pl}(0) = 140\text{mm}$). The desired pitch angle trajectory was set as:

$$\theta_d = \begin{cases} -15^\circ, & t \in [0, 100]\text{s}, \\ -30^\circ, & t \in (100, 200]\text{s}. \end{cases} \quad (67)$$

The experimental tracking results under these conditions for the three control strategies (SMC, TSMC, FSMC) are illustrated in Fig. 15.

The pitch angle convergence performance of the three control methods under different initial states is shown in Table 8. From the results, it is evident that the controllers exhibit markedly different convergence behaviors. The SMC strategy shows pronounced chattering, with convergence times ranging from approximately 55 s to 87 s across all initial conditions. Moreover, when the reference signal switches from -15° to -30° , the convergence of SMC lags behind the other two methods. In

Table 8

Pitch angle convergence time of three control methods under different initial states in Case 5.

Initial state	Pitch angle convergence time(s)		
	SMC	TSMC	FSMC
-8°	55	42	50
5°	68	57	51
10°	87	75	52

addition, the pitch angle tracking error using the SMC control method is larger than that of TSMC and FSMC. Although TSMC achieves relatively faster convergence than SMC, it suffers from significant oscillations around the target state. Additionally, its convergence time is sensitive to the magnitude of the initial tracking error. Specifically, for initial pitch angles of -8° , 5° , and 10° , the convergence times are 42 s, 57 s, and 75 s, respectively, indicating a clear dependence on initial conditions.

In contrast, the FSMC scheme demonstrates strong robustness to initial state variations. As shown in Fig. 15(c), its convergence time remains nearly constant at approximately 50 s across all test scenarios. Fig. 16 shows the UG prototype achieving a commanded pitch angle change from -15° to -30° in the pool experiment via the FSMC algorithm. Fig. 17 shows the comparison results of the pitch angle tracking errors of SMC, TSMC and FSMC controllers under different initial conditions of UG. As shown in Fig. 17(a), the convergence error of FSMC remains at around 3° , which is smaller than that of SMC and TSMC.

This validates the fixed-time convergence property of FSMC and highlights its superior performance in achieving rapid and consistent response irrespective of initial conditions.

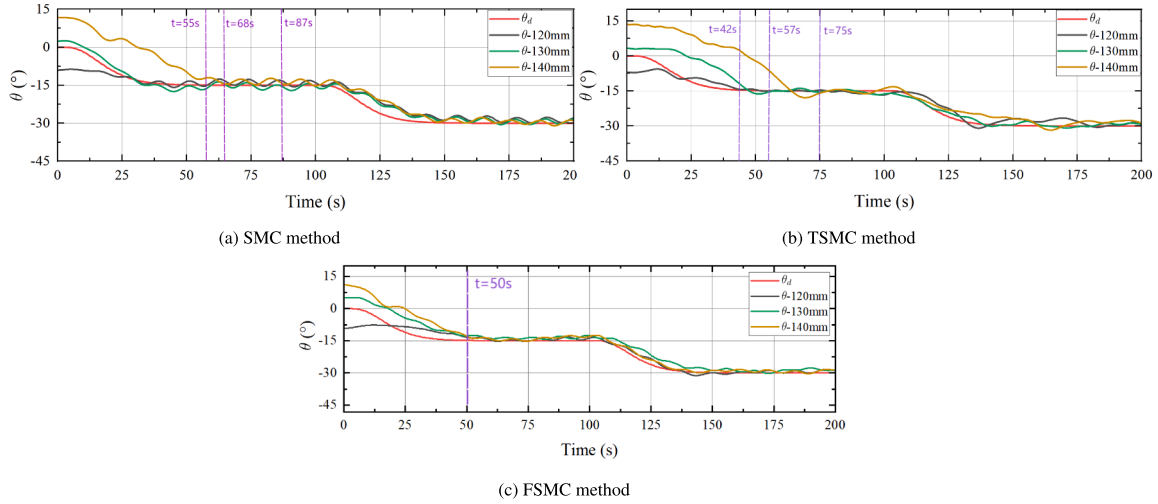


Fig. 15. Comparison of pitch angle tracking results for SMC, TSMC, and FSMC controllers under varying initial conditions.

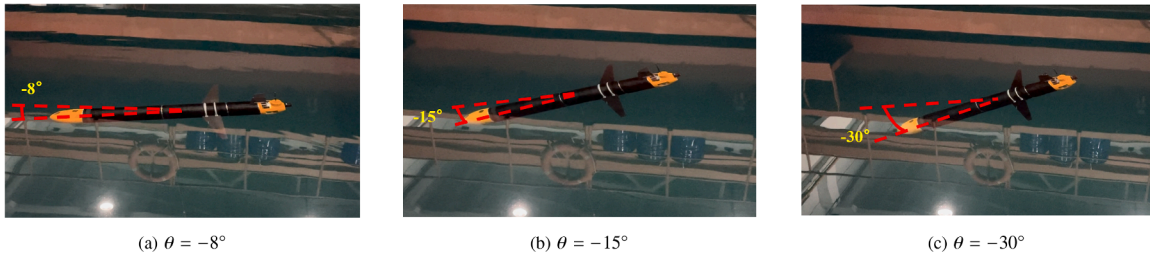


Fig. 16. When the initial pitch angle of the UG prototype is -8° , FSMC is used to track the desired pitch angle.

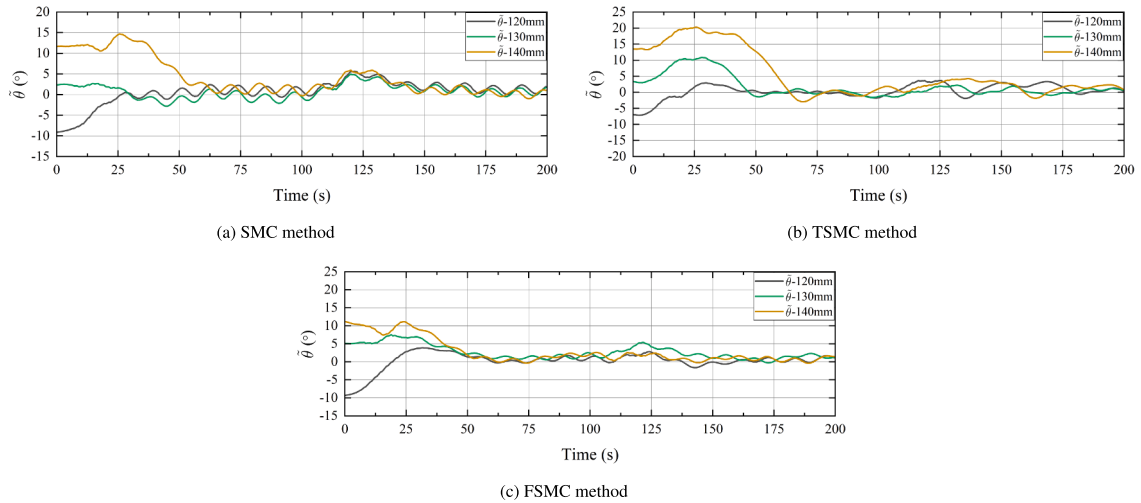


Fig. 17. Comparison of pitch angle tracking error results of SMC, TSMC, and FSMC controllers under different initial conditions.

5. Conclusion and discussion

This study investigated the trajectory tracking control of an underwater glider (UG) under constrained task scenarios. A fixed-time non-singular terminal sliding mode control (FSMC) strategy was proposed for trajectory tracking in the vertical plane. In addition, a dual-loop pitch angle tracking control architecture was developed to enable efficient implementation on the STM32-based main controller of the UG prototype. Numerical simulation results demonstrated that the proposed FSMC-based tracking method achieved faster convergence and smaller

steady-state errors compared to the conventional sliding mode control (SMC) scheme. Furthermore, compared with the terminal sliding mode control (TSMC) method, the FSMC scheme exhibited convergence behavior that is independent of the initial conditions. Experimental validation using the UG prototype further confirmed the effectiveness of the FSMC-based control strategy. Specifically, in slider tracking experiments, the FSMC controller achieved convergence within approximately 28 s, outperforming both SMC and TSMC approaches. In addition, pitch angle tracking tests conducted in a laboratory pool environment showed that the FSMC algorithm maintained its fixed-time convergence

characteristics and robustness with respect to varying initial pitch angles, and consistently stabilizing the pitch angle error within approximately 3° .

In future work, the proposed fixed-time control method will be integrated with path planning strategies tailored to constrained observation missions. Particular emphasis will be placed on the development of adaptive thermocline tracking algorithms. Due to the limitations of the laboratory pool facilities, verification of the UG trajectory tracking performance using the FSMC algorithm under external wave disturbances was not conducted. However, field experiments using the UG prototype will be carried out to further evaluate the proposed planning and control framework under realistic marine environmental conditions. These findings contribute to the advancement of robust and time efficient control strategies for UGs engaged in boundary constrained ocean observation missions.

CRedit authorship contribution statement

Xu Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Yu Cao:** Visualization, Investigation, Data curation; **Hexiong Zhou:** Writing – review & editing, Supervision, Software, Funding acquisition, Formal analysis, Data curation; **Baoheng Yao:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis; **Lian Lian:** Resources, Funding acquisition; **Zhihua Mao:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The present work is supported by [National Key R&D Program of China \(2023YFC2811103, 2021YFC2801600\)](#), [Natural Science Foundation of China](#) under Grant Number 42406183, the Oceanic Interdisciplinary Program of Shanghai Jiao Tong University under Grant SL2022ZD206, and Scientific Research Fund of Second Institute of Oceanography, MNR under Grant SL2302.

References

- Cao, J., Cao, J., Zeng, Z., Lian, L., 2016. Nonlinear multiple-input-multiple-output adaptive backstepping control of underwater glider systems. *Int. J. Adv. Rob. Syst.* 13 (6). <https://doi.org/10.1177/1729881416669484>
- Cao, J., Lu, D., Li, D., Zeng, Z., Yao, B., Lian, L., 2019. Smartfloat: a multimodal underwater vehicle combining float and glider capabilities. *IEEE Access* 7, 77825–77838.
- Chen, H., Tang, G., Wang, S., Guo, W., Huang, H., 2023. Adaptive fixed-time backstepping control for three-dimensional trajectory tracking of underactuated autonomous underwater vehicles. *Ocean Eng.* 275, 114109.
- Feng, H., Yu, J., Huang, Y., Qiao, J., Wang, Z., Xie, Z., Liu, K., 2021. Adaptive coverage sampling of thermocline with an autonomous underwater vehicle. *Ocean Eng.* 233, 109151. <https://doi.org/10.1016/j.oceaneng.2021.109151>
- Feng, Y., Yu, X., Han, F., 2013. On nonsingular terminal sliding-mode control of nonlinear systems. *Automatica* 49 (6), 1715–1722.
- Feng, Y., Yu, X., Man, Z., 2002. Non-singular terminal sliding mode control of rigid manipulators. *Automatica* 38 (12), 2159–2167.
- Joo, M., Qu, Z.H., 2015. An autonomous underwater vehicle as an underwater glider and its depth control. *Int. J. Contr. Autom. Syst.* 13 (5), 1212–1220. <https://doi.org/10.1007/s12555-014-0252-8>
- Leonard, N.E., Graver, J.G., 2001. Model-based feedback control of autonomous underwater gliders. *IEEE J. Oceanic Eng.* 26 (4), 633–645. <https://doi.org/10.1109/48.972106>
- Leonard, N.E., Paley, D.A., Davis, R.E., Fratantoni, D.M., Lekien, F., Zhang, F., 2010. Co-ordinated control of an underwater glider fleet in an adaptive ocean sampling field experiment in monterey bay. *J. Field Rob.* 27 (6), 718–740.
- Liu, X., Jia, C., Wang, H., 2025. Practical variable exponent fixed-time nonsingular sliding mode control for nonlinear systems and its applications. *IEEE Trans. Autom. Sci. Eng.*, 22, 10871–10880.
- Mahmoudian, N., Woolsey, C., 2008. Underwater glider motion control. In: 2008 47th IEEE Conference on Decision and Control, pp. 552–557. <https://doi.org/10.1109/CDC.2008.4739432>
- Ni, J., Liu, L., Liu, C., Hu, X., Li, S., 2016. Fast fixed-time nonsingular terminal sliding mode control and its application to chaos suppression in power system. *IEEE Trans. Circuits Syst. II Express Briefs* 64 (2), 151–155.
- Noh, M.M., Arshad, M.R., Mokhtar, R.M., 2011. Depth and pitch control of USM underwater glider: performance comparison PID vs. LQR. *Indian J. Mar. Sci.* 40, 200–206. <http://nopr.niscares.in/handle/123456789/11724>
- Paley, D.A., Zhang, F., Leonard, N.E., 2008. Cooperative control for ocean sampling: the glider coordinated control system. *IEEE Trans. Control Syst. Technol.* 16 (4), 735–744.
- Polyakov, A., 2011. Nonlinear feedback design for fixed-time stabilization of linear control systems. *IEEE Trans. Automat. Contr.* 57 (8), 2106–2110.
- Polycarpou, M.M., 1996. Stable adaptive neural control scheme for nonlinear systems. *IEEE Trans. Automat. Contr.* 41 (3), 447–451.
- Shtessel, Y., Edwards, C., Fridman, L., Levant, A., et al., 2014. *Sliding Mode Control and Observation*. 10. Springer.
- Su, B., Wang, H., Wang, Y., Gao, J., 2021. Fixed-time formation of AUVs with disturbance via event-triggered control. *Int. J. Control Autom. Syst.* 19 (4), 1505–1518.
- Su, Z.-q., Zhou, M., Han, F.-f., Zhu, Y.-w., Song, D.-l., Guo, T.-t., 2019. Attitude control of underwater glider combined reinforcement learning with active disturbance rejection control. *J. Mar. Sci. Technol.* 24(3), 686–704.
- Sun, H., Zong, G., Cui, J., Shi, K., 2022. Fixed-time sliding mode output feedback tracking control for autonomous underwater vehicle with prescribed performance constraint. *Ocean Eng.* 247, 110673.
- Tatone, F., Vaccarini, M., Longhi, S., 2009. Modeling and attitude control of an autonomous underwater glider. *IFAC Proc. Vol.* 42 (18), 217–222.
- Wan, L., Zhang, D., Sun, Y., Qin, H., Cao, Y., Chen, G., 2022. Fast fixed-time vertical plane motion control of autonomous underwater gliders in shallow water. *J. Franklin Inst.* 359 (18), 10483–10509.
- Wang, G., Yu, J., Yang, Y., 2025. Enhancing trajectory tracking performance of underwater gliders using finite-time sliding mode control within a reinforcement learning framework. *J. Mar. Sci. Eng.* 13 (5), 884.
- Wang, J., Wu, Z., Tan, M., Yu, J., 2021. Model predictive control-based depth control in gliding motion of a gliding robotic dolphin. *IEEE Trans. Syst. Man Cybern. Syst.* 51 (9), 5466–5477. <https://doi.org/10.1109/TSMC.2019.2956531>
- Yang, C., Peng, S., Fan, S., Zhang, S., Wang, P., Chen, Y., 2016. Study on docking guidance algorithm for hybrid underwater glider in currents. *Ocean Eng.* 125, 170–181.
- Yang, H., Ma, J., 2010. Nonlinear control for autonomous underwater glider motion based on inverse system method. *J. Shanghai Jiaotong Univ.* 15, 713–718.
- Yu, X., Feng, Y., Man, Z., 2020. Terminal sliding mode control—an overview. *IEEE Open J. Indus. Electron. Soc.* 2, 36–52.
- Zak, M., 1988. Terminal attractors for addressable memory in neural networks. *Phys. Lett. A* 133 (1–2), 18–22.
- Zhang, X., Yao, B., Lian, L., Mao, Z., 2024. Adaptive neural network sliding mode tracking control with prescribed performance for an underwater glider under input saturation. *Ocean Eng.* 307, 118150.
- Zhang, X., Zhou, H., Fu, J., Wen, H., Yao, B., Lian, L., 2023. Adaptive integral terminal sliding mode based trajectory tracking control of underwater glider. *Ocean Eng.* 269, 113436.
- Zhou, H., Wei, Z., Zeng, Z., Yu, C., Yao, B., Lian, L., 2020. Adaptive robust sliding mode control of autonomous underwater glider with input constraints for persistent virtual mooring. *Appl. Ocean Res.* 95, 102027.
- Zhu, Z., Duan, Z., Qin, H., Xue, Y., 2023. Adaptive neural network fixed-time sliding mode control for trajectory tracking of underwater vehicle. *Ocean Eng.* 287, 115864. <https://doi.org/10.1016/j.oceaneng.2023.115864>
- Zou, H., Zhang, G., Hao, J., 2024. Nonsingular fast terminal sliding mode tracking control for underwater glider with actuator physical constraints. *ISA Trans.* 146, 249–262.
- Zuo, Z., Han, Q.-L., Ning, B., Ge, X., Zhang, X.-M., 2018. An overview of recent advances in fixed-time cooperative control of multiagent systems. *IEEE Trans. Ind. Inf.* 14 (6), 2322–2334.